

Lab 10: Atomic Absorption Spectroscopy

Chemistry 150 - Winter 2026

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Data

Table 1. Raw measurements.

Measurement	Value
Mass of copper metal (g)	0.126

Table 2. Absorbance (signal) readings from AAS instrument.

Sample	Volume Pipetted (mL)	Absorbance
Standard 1	1	0.014
Standard 2	2	0.026
Standard 3	5	0.067
Standard 4	10	0.134
Fountain (unknown)	N/A	0.003

Calculations

Standard Concentrations

The stock solution was prepared by dissolving 0.126 g of copper in a 100 mL volumetric flask:

$$C_1 = \frac{0.126 \text{ g}}{0.100 \text{ L}} = 1.26 \text{ g/L} = 1260 \text{ mg/L}$$

First dilution: 10 mL of stock into a 100 mL volumetric flask:

$$C_2 = 1260 \times \frac{10}{100} = 126 \text{ mg/L}$$

Second dilution: 10 mL of the above into a 100 mL volumetric flask:

$$C_3 = 126 \times \frac{10}{100} = 12.6 \text{ mg/L}$$

Working standards were prepared by pipetting V_n mL of the 12.6 mg/L solution into 50 mL volumetric flasks:

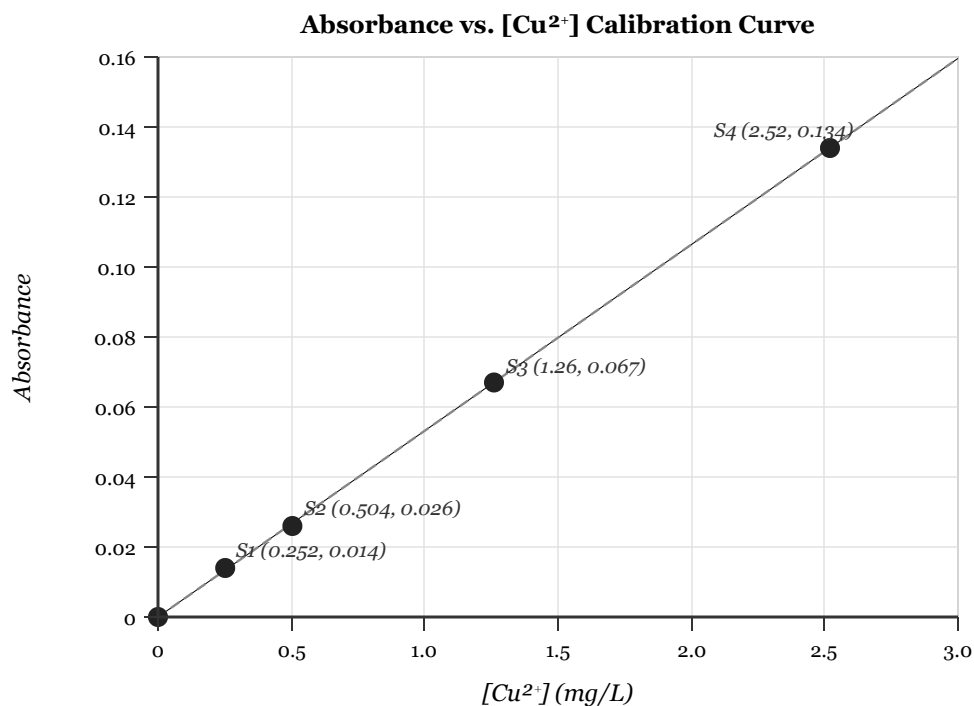
$$C_n = 12.6 \times \frac{V_n}{50} \text{ mg/L}$$

Table 3. Calculated concentrations and measured absorbances for each standard.

Standard	Volume (mL)	Concentration (mg/L)	Absorbance
S1	1	0.252	0.014
S2	2	0.504	0.026
S3	5	1.26	0.067
S4	10	2.52	0.134

Calibration Curve

$$A = 0.0532 [\text{Cu}^{2+}] \quad (R^2 = 0.9999)$$



Concentration of Fountain Water

Solving for $[\text{Cu}^{2+}]$ using the measured fountain absorbance of 0.003:

$$[\text{Cu}^{2+}] = \frac{A}{0.0532} = \frac{0.003}{0.0532} = 0.056 \approx 0.06 \text{ mg/L}$$

Conclusion

The copper concentration in the drinking fountain water was determined to be approximately **0.06 mg/L**.